

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

6th Semester of B. Pharm. Examination

University Theory Examination April/May 2015

PH313 BIOPHARMACEUTICS & PHARMACOKINETICS

Date: 30.04.15, Thursday

Time: 10.00 a.m to 1.00 p.m

Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

Q 1 Attempt any TWO of the following

- A** Briefly describe the process of drug distribution in the body and write down (10)
factors determining distribution of drugs.
- B** I. What are various mechanism for drug transport in body? (05)
II. Explain the detail concept of volume of distribution and clearance. (05)
- C** What are the importance of Phase 1 and Phase 2 reactions? Explain in brief each (10)
of them by giving examples.

Q 2 Attempt any TWO of the following

- A** Enlist out factors affecting Drug Absorption and write in detail about (10)
Physicochemical factors affecting Drug Absorption.
- B** List out principal organs involved in Drug Excretion and explain in detail about (10)
renal and biliary drug excretion process.
- C** I. Draw a labelled diagram of plasma concentration vs. Time Profile and (6)
explain following terms.
a. Therapeutic Window
b. T_{max}
c. Onset of Action
d. Duration of Action
e. Maximum Safe concentration
f. Minimum Effective Concentration
II. Explain detail about First Pass Metabolism. (4)

SECTION – II

Q.3 Attempt **any TWO** of the following:

- A) I. Write a short note on "Wagner Nelson Method". (05)
II. Explain Loo-Riegelman method with its applications. (05)
- B) Define Bioavailability. Explain methods for determining bioavailability of drugs. (10)
- C) I. Explain One compartment open model extravascular administration. (05)
II. Give one example each for ADME where nonlinear pharmacokinetics is observed. (05)

Q.4 Attempt **any TWO** of the following:

- A) I. Define following terms. (05)
 - a. Bioequivalence
 - b. Extraction Ratio
 - c. Therapeutic Equivalence
 - d. Clinical Pharmacokinetics
 - e. Pharmacokinetic ModelsII. Discuss various factors affecting BA of drugs. (05)
- B) I. Write a short note on curve fitting method. (05)
II. Explain the methods to determine K_m and V_{max} for a drug exhibiting non-linear pharmacokinetics. (05)
- C) I. What are the advantages and disadvantages of compartmental models? (04)
II. Write a brief note on IVIVC and Biological classification system (BCS Classification). (06)